Essays

|!|BCEEC0120000001|!|
Describe the events that occur during one complete cell cycle in an animal cell.

(13 marks)

##

Interphase (Max. 3)

During interphase, the cell carries out the following activities to prepare for the next cell division: (1)

- DNA replication (1)
- Synthesis of proteins (1)
- Synthesis of new organelles (1)

Mitotic cell division (Max. 7)

- Prophase: The chromatin / replicated DNA coils up tightly into chromosomes. (1)
  The nuclear membrane disappears. (1)
- Metaphase: The chromosomes line up along the equatorial plane. (1)
  Spindle fibres attach to the chromosomes at the centromeres. (1)
- Anaphase: The spindle fibres shorten and pull the sister chromatids of each chromosomes towards the opposite poles of the cell. (1)
- Telophase: A nuclear membrane forms around each group of daughter chromosomes. (1)
  The chromosomes loosen to become chromatin again. (1)
- Cytokinesis: The cell membrane around the middle of the dividing cell constricts to form a cleavage furrow. The furrow further deepens to divide the cytoplasm into two halves. (1)

Communication (Max. 3)

(13 marks)

##

Skill assessed	Marks
Understanding basic concepts	10
Communication	3

© 2020 Aristo Educational Press Ltd.

|!|BCEEC0120000002|!|

Discuss meiotic cell division and its importance to organisms. (10 marks)

##

Content	Descriptor	Marks
1. Introduction of meiotic cell division	- It is a kind of cell division which involves two successive divisions: meiosis I and meiosis II. - It is found in the gamete-producing cells found in the sex organs (e.g. anthers and ovaries in flowering plants; testis and ovaries in animals). - In meiosis, pairs of homologous chromosomes line up and arrange themselves randomly in the middle of the cell. - Each homologous chromosomes pair separates independently and moves to opposite pole of cells. - Each cell produces four haploid cells per meiosis.	(Max. 4)
2. Importance to organisms	- It forms haploid gametes in organisms for sexual reproduction by halving the chromosome number. - The independent assortment and crossing over of homologous chromosomes produce new genetic combinations in gametes. - At fertilisation, the random fusion of gametes further increases genetic variations among the offspring.	(Max. 3)
Communication		(Max. 3)

(10 marks)

##

Skill assessed	Marks
Understanding basic concepts	7
Communication	3

© 2020 Aristo Educational Press Ltd.